GEOMETRIC IRREDUCIBILITY IS ZARISKI LOCAL MODULO CONNECTEDNESS: FOUR EQUIVALENT FORMULATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the chapter's slogan environment, including the author's spelling of 'irreductibility'; it is reproduced verbatim and rendered by the no-op display environment of the retained preamble.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-irreducible-local is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Geometric irreducibility of a k-algebra. Algebra, Definition algebra-definition-geometrically-irreducible, invoked at proof lines 1413-1415 and 1429-1431. Supplies the affine translation used at the start of the proof.
- Irreducibility of a scheme tested on affine opens. Properties, Lemma properties-lemma-characterize-irreducible, invoked at proof lines 1434-1436. Supplies that a scheme with irreducible affine opens is irreducible.
- A cover by pairwise intersecting irreducible opens makes a space irreducible. Standard topology, used at proof lines 1439-1442. Supplies the last implication, applied after base change.

Source reference. Original definition 037L: sources/stacks-chow/tags.txt, source lines 4204–4204. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OM: sources/stacks-chow/tags.txt, source lines 2199–2199. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Geometric irreductibility is Zariski local modulo connectedness. Let k be a field. Let X be a scheme over k. The following are equivalent*

(1) *X is geometrically irreducible over k,*
(2) *for every nonempty affine open U the k-algebra $\mathcal{O}_X(U)$ is geometrically irreducible over k (see Algebra, Definition 037L),*
(3) *X is irreducible and there exists an affine open covering $X = \bigcup U_i$ such that each k-algebra $\mathcal{O}_X(U_i)$ is geometrically irreducible, and*

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(4) *there exists an open covering* $X = \bigcup_{i \in I} X_i$ *with* $I \neq \emptyset$ *such that* X_i *is geometrically irreducible for each* i *and such that* $X_i \cap X_j \neq \emptyset$ *for all* $i, j \in I$.

Moreover, if X *is geometrically irreducible so is every nonempty open subscheme of* X.

Proof. An affine scheme $\mathrm{Spec}(A)$ over k is geometrically irreducible if and only if A is geometrically irreducible over k; this is immediate from the definitions. Recall that if a scheme is irreducible so is every nonempty open subscheme of X, any two nonempty open subsets have a nonempty intersection. Also, if every affine open is irreducible then the scheme is irreducible, see Properties, Lemma 01OM. Hence the final statement of the lemma is clear, as well as the implications (1) $\Rightarrow$ (2), (2) $\Rightarrow$ (3), and (3) $\Rightarrow$ (4). If (4) holds, then for any field extension k'/k the scheme $X_{k'}$ has a covering by irreducible opens which pairwise intersect. Hence $X_{k'}$ is irreducible. Hence (4) implies (1). $\square$